

Electrostatics & Magnetostatics

$$E. \quad \begin{cases} \nabla \cdot \vec{E} = \rho / \epsilon_0 & \text{'Gauss's law'} \\ \nabla \times \vec{E} = 0 \end{cases}$$

$$\left. \begin{aligned} \nabla \cdot \vec{B} &= 0 \\ \nabla \times \vec{B} &= \mu_0 \vec{J} \end{aligned} \right\} M.$$

$$\vec{F} = q (\vec{E} + \vec{v} \times \vec{B}) \quad L. \text{ Force}$$

vector potⁿ $\vec{A}$, $\nabla \cdot \vec{B} = 0 \Rightarrow \nabla \cdot (\nabla \times \vec{A}) = 0$
 $\nabla \cdot (\nabla \times (\vec{A} + \nabla \phi)) = 0$

$$\nabla \times (\nabla \phi) = 0$$

$\vec{A}$ can't be unique.

$$\nabla \cdot \vec{B} = \rho / \epsilon_0 \rightarrow \rho \rightarrow \text{presence of monopoles.}$$

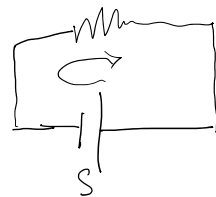
Electromotive force

$$\vec{f} = \vec{f}_s + \vec{E}$$

$$\underline{E} = \oint \vec{f} \cdot d\vec{l}$$

$$= \oint \vec{f}_s \cdot d\vec{l}$$

$$\text{b. } \oint \vec{E} \cdot d\vec{l} = 0$$



emf is the integral of a force per unit charge.

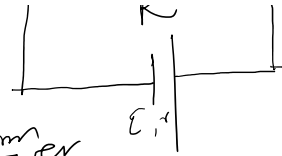
Q. Prob.
7.5

emf $\mathcal{E}$, internal resistance r , load R .

$$i = \frac{V}{R} = \frac{\mathcal{E}}{r + R}$$



$$i = \frac{V}{R} = \frac{\mathcal{E}}{r+R}$$



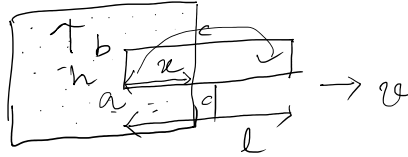
$$P = i^2 R$$

maximum power

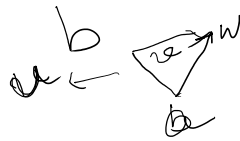
$$\frac{dP}{dR} = 0$$

$$\Rightarrow R = r$$

Motional emf
Generator



$$\mathcal{E} = \oint \vec{F}_{mag} \cdot d\vec{l} = \underline{v B h}$$



$$\mathcal{E} = \underline{v B h}$$

$$\phi = \int B \cdot d\vec{a}$$

ϕ is the flux through the loop.

Conductivity

$$\underline{\underline{\vec{J}}} = \sigma \underline{\underline{\vec{E}}}$$

σ conductivity

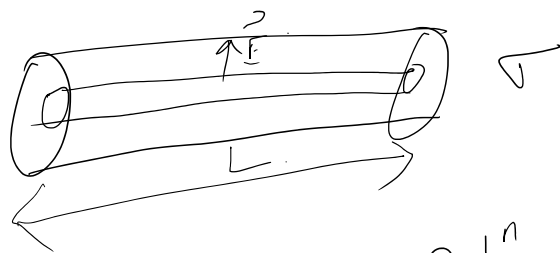
$$\left(\vec{J} \propto \vec{E} \Rightarrow \vec{J} \propto \vec{F} \right)$$

understand

Resistivity $(R)\rho = 1/\sigma$

coaxial cable

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{s}$$



$$I = \frac{2\pi\sigma L}{\ln(b/a)} V$$

V - potⁿ diff. created by electric field $\vec{E}$.

$$\boxed{I \propto V} \text{ Ohm's law, } V = IR$$

$$R = \left(\frac{2\pi\sigma L}{\ln(b/a)} \right)^{-1}$$

EMF

09-09-24

Force per unit charge in a circuit,

$$\mathcal{F} = \mathcal{F}_s + E$$

integral of, $\mathcal{E} = \oint \mathcal{F} \cdot d\mathbf{l} = \oint \mathcal{F}_s \cdot d\mathbf{l}$

$$V = \oint_a^b \mathcal{F}_s \cdot d\mathbf{l} = \oint \mathcal{F}_s \cdot d\mathbf{l} = \mathcal{E}$$

$\mathcal{F}_s = 0$
for a ideal

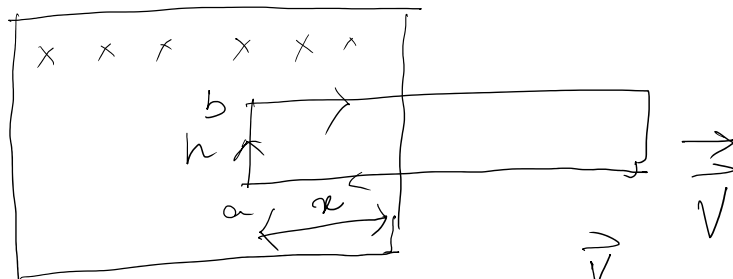
$$E + \mathcal{F}_s = 0$$

$$\Rightarrow E = -\mathcal{F}_s$$

$$\parallel$$

$$= - \int E \cdot d\mathbf{l}$$

$\mathcal{E} \equiv$ work done per unit charge.



$$\mathcal{E} = Bbh$$

$$\mathcal{E} = Bbh = \oint \mathcal{F}_s \cdot d\mathbf{l}$$

$$= BW h \sin\theta$$

$$= \oint B \frac{u}{\cos\theta} h \sin\theta = \underline{\underline{Bbh}}$$

magnetic
flux

$$\phi = \oint_s B \cdot d\mathbf{a} = 0$$

$$\phi = Bhx$$

$$\frac{d\phi}{dt} = B h \frac{dx}{dt} = -Bbhv$$

$$\Phi = B h x$$

$$\frac{d\Phi}{dt} = \frac{d}{dt} (B h x) = B h \frac{dx}{dt} = -B h v$$

$$\boxed{\mathcal{E} = - \frac{d\Phi}{dt}} \quad \text{Flux Rule}$$

Motional emf

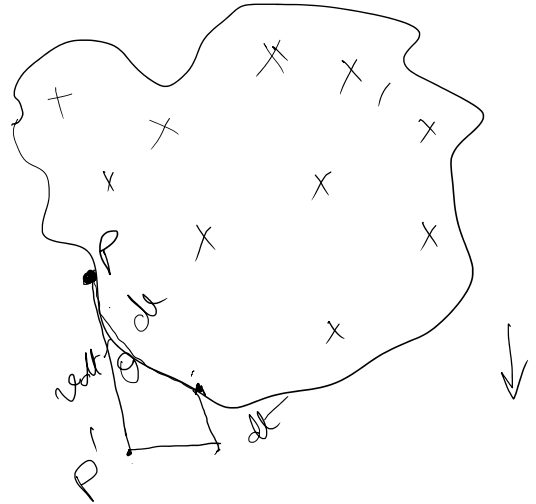
$$\int d\Phi = \int \vec{B} \cdot d\vec{a}$$

$$\vec{w} = \vec{v} + \vec{u}$$

$$da = (\vec{v} \times d\vec{l}) dt$$

$$\frac{d\Phi}{dt} = \oint \vec{B} \cdot (\vec{v} \times d\vec{l})$$

$$= \oint \vec{B} \cdot (\vec{w} \times d\vec{l})$$



Scalar triple product

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = - \vec{B} \cdot (\vec{A} \times \vec{C}) = - \vec{C} \cdot (\vec{B} \times \vec{A})$$

$$\vec{B} \cdot (\vec{w} \times d\vec{l}) = - (\vec{w} \times \vec{B}) \cdot d\vec{l}$$

$$\frac{d\Phi}{dt} = - \int (\vec{w} \times \vec{B}) \cdot d\vec{l}$$

$$= - \oint \vec{f}_{\text{mag}} \cdot d\vec{l} = - \mathcal{E}$$

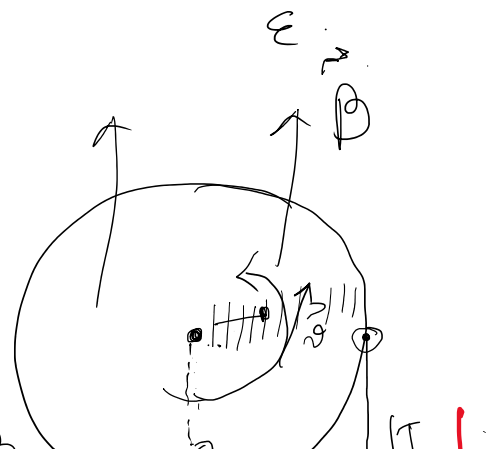
A changing mag. field induces electric field.

$$\vec{f} \propto \vec{v} \times \vec{B}$$

$$\vec{f}_{\text{mag}} = \vec{v} \times \vec{B}$$

$$\vec{v} = \omega \times \vec{r}$$

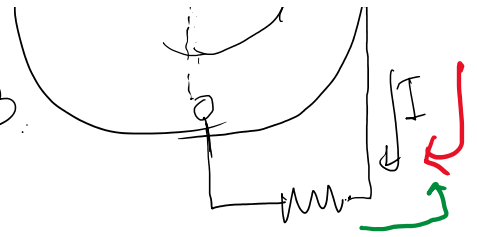
$$\mathcal{E} = \int \vec{f}_{\text{mag}} \cdot d\vec{r}$$



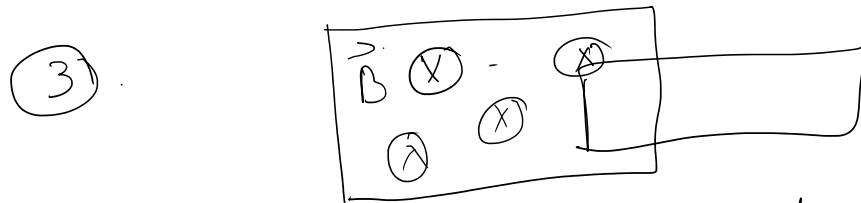
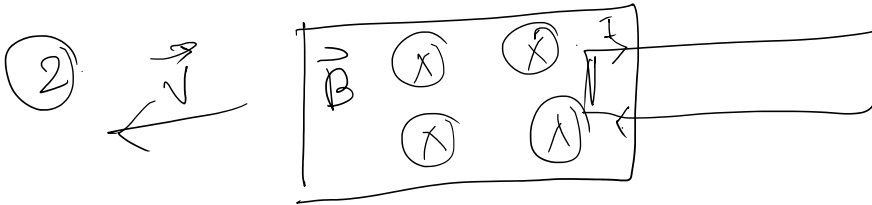
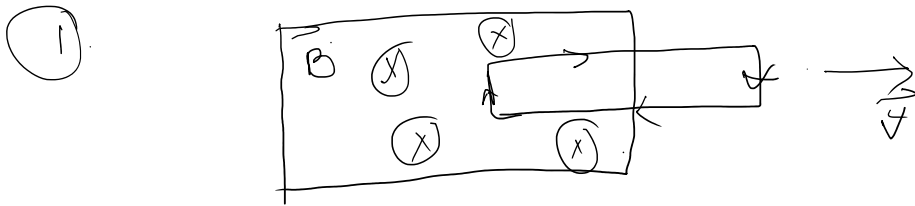
$$\mathcal{E} = \int \vec{f}_{\text{mag}} \cdot d\vec{s}$$

$$= \int \omega \times \vec{B} \cdot d\vec{r} = \frac{\omega a^2 B}{2}$$

$$I = \frac{\mathcal{E}}{R} = \frac{\omega B a^2}{2R}$$



Faraday's Experiment



$$\mathcal{E} = - \frac{d\phi}{dt} = - \frac{d}{dt} \int \vec{B} \cdot d\vec{a}$$

$$= - \int \vec{B} \cdot \frac{d\vec{a}}{dt}$$

$$= - \int \frac{d\vec{B}}{dt} \cdot d\vec{a}$$

$$\mathcal{E} = \oint \vec{f}_{\text{mag}} \cdot d\vec{l}$$